

Generative Learning for Ambisonic Upscaling

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Abstract—Ambisonics upscaling (AU) aims to enhance the spatial resolution of sound fields by estimating high-order Ambisonics (HOA) components from low-order observations. While deep learning and model-based strategies have been considered for AU, both approaches exhibit significant performance degradation in realistic scenarios, where reverberant sound fields violate the directional sparsity inherent to discriminative mappings. In this work, we address AU as a generative task rather than a deterministic reconstruction, expanding generative modeling to specifically target the recovery of spatial information in reverberant speech. We investigate two dominant continuous-time generative paradigms, adapting both Score-based Generative Model and Flow Matching to these complex acoustic settings. We provide an extensive numerical study comparing our methods against state-of-the-art baselines in various acoustic scenarios. Additionally, we conduct subjective listening tests to evaluate the perceived quality and spatial accuracy of the proposed generative framework across various reverberant scenarios. The studies reveal that Flow Matching consistently outperforms both its discriminative counterparts and Diffusion-based paradigms in all reverberant settings.

I. INTRODUCTION

Spatial audio technology aims to reproduce sound fields in a way that preserves the spatial cues perceived by human listeners, such as the direction and distance of sound sources [2]. By reconstructing the acoustic scene in three dimensions, spatial audio enables immersive auditory experiences and is widely used in applications including virtual and augmented reality, gaming, cinema production, teleconferencing, and music reproduction [3]. Among spatial audio representations, Ambisonics has emerged as a particularly attractive framework due to its flexibility in capturing, encoding, and rendering sound fields [4]. Ambisonics represents the sound field using spherical harmonic basis functions, allowing the representation to remain independent of the recording and playback systems and enabling scalable spatial resolution.

In Ambisonics, the spatial resolution of the sound field is determined by the order of the spherical harmonic expansion. First-order Ambisonics (FOA) employs four channels corresponding to the zeroth- and first-order harmonics, which can be captured using relatively compact and affordable microphone arrays [5]. However, due to the limited number of spatial components, FOA provides only coarse spatial resolution, which can degrade localization accuracy and reduce the sense of immersion. Increasing the Ambisonic order to high-order Ambisonics (HOA) representations that include additional spherical harmonic components allows capturing finer spatial details [6]. The drawback is that

acquiring HOA signals typically requires large and costly microphone arrays with many sensors, which restricts their practical deployment. Consequently, there is a growing interest in developing *Ambisonics upscaling (AU)* techniques that estimate high-order components from low-order recordings, enabling enhanced spatial resolution without requiring complex acquisition hardware.

Early approaches to AU are predominantly model-based and rely on physical models of the sound field. A common strategy is to perform plane wave decomposition (PWD) and reconstruct the missing high-order components by exploiting sparsity assumptions on the spatial distribution of sources. In particular, several works formulate the problem using compressed sensing (CS), where the sound field is assumed to consist of a small number of plane waves arriving from distinct directions [7]–[11]. Under this assumption, the HOA coefficients can be inferred by solving a sparse recovery problem [12]. Alternatively, parametric methods bypass explicit spatial sparsity assumptions by directly estimating the sound field’s parameters and resynthesizing the sound field to a given order [13], [14]. While such techniques can yield accurate reconstructions under ideal free-field conditions with limited noise and few active sources, their performance degrades significantly in challenging acoustic environments when the underlying assumptions do not hold.

Motivated by the success of deep learning in natural language processing and computer vision, recent studies have explored data-driven paradigms for spatial audio enhancement, spanning both discriminative and generative frameworks. Representative discriminative approaches include the study in [15], which proposed a transformer-based architecture to synthesize Ambisonic signals from stereo inputs, and the usage of physically informed neural networks suggested in [16] to upscale spherical microphone array recordings by enforcing wave-equation consistency. In parallel, the emergence of generative modeling has introduced new capabilities for spatial synthesis. Early efforts in this direction, such as the work by Xia *et al.* [17], utilized Generative Adversarial Networks (GANs) to upscale FOA Room Impulse Responses (RIRs), aiming to recover high-resolution spatial characteristics from room impulse responses. Building on this, Picard *et al.* [18] employed diffusion models to upscale the steering vectors of FOA signals, while Kushwaha *et al.* [19] demonstrated the generation of FOA content directly from textual descriptions. Furthermore, Liang *et al.* [20] utilized Flow Matching (FM) to synthesize 7.1.4 surround sound layouts from stereo sources.

Specifically for the task of AU, mostly discriminative approaches have been introduced. For instance, Gao *et al.* [21] proposed a multi-scale convolutional neural network (CNN) operating in the frequency domain with sparse encoding, while Routray *et al.* [22] presented a multi-stage deep neural network (DNN) architecture to upsample signals order-by-order. Recently, Chatzimoustafa *et al.* [23] refined this hierarchical approach by replacing linear layers with gated recurrent units

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(GRUs) to better capture temporal dependencies. However, these models, including more recent encoder-decoder architectures like adapted Conv-TasNet (CTN) [24], primarily demonstrate success in anechoic or low reverberation conditions. Generative modeling has to date only been considered for AU in the work from 2021 by Zhang *et al.* [25], which explored the use of early generative architectures based on GANs for this task, focusing on single-tone signals from specific directions.

Despite these advances in AU, a major challenge remains the reconstruction of spatial details in reverberant environments. In such scenarios, late reflections generate spatially complex, diffuse-like sound fields that deviate significantly from the directional sparsity assumptions underlying many model-based techniques [26]. At the same time, discriminative learning methods that rely on deterministic mappings [27] from low- to high-order representations often struggle to capture the complex and stochastic spatial structure introduced by reverberation. As a result, existing methods frequently produce spatially blurred reconstructions in which distinct directional cues are not well resolved, leading to a loss of high-frequency spatial detail and reduced perceptual immersion.

To address the limitations of existing approaches, we introduce a generative framework for AU based on state-of-the-art *continuous-time generative modeling*. Our approach casts AU as sampling from the conditional distribution of missing high-order spatial components given the observed low-order Ambisonic signals. Unlike conventional approaches that rely on deterministic mappings from low- to high-order representations, our formulation treats the reconstruction of high-order harmonics as a generative inference problem, allowing the model to capture the stochastic spatial structures that arise in reverberant sound fields. To extend AU into a generative paradigm for challenging acoustic conditions, we investigate continuous-time generative formulations that provide improved training stability and sampling efficiency. The proposed framework combines flexible architectural design with physically motivated reconstruction constraints to generate spatially consistent HOA representations while preserving the reliability of the measured low-order components.

Our main contributions are summarized as follows:

- **Generative formulation for AU:** We introduce a generative perspective for AU, in which the missing HOA components are recovered by modeling their conditional distribution given the observed low-order signals. This formulation enables the reconstruction process to account for the inherent ambiguity and stochasticity of spatial sound fields in reverberant environments.
- **Continuous-time generative modeling for spatial audio:** We investigate two dominant continuous-time generative paradigms for the AU task, namely Score-based Generative Model (SGM) [28] and FM [29]. While SGM reconstructs the target distribution through reverse diffusion driven by score estimation, FM learns a deterministic vector field that transports samples from a noise prior to the target HOA distribution, offering improved stability and sampling efficiency.
- **Targeted reconstruction of high-order spatial components:** We introduce a reconstruction strategy that preserves the measured low-order Ambisonic components while re-

stricting the generative model to synthesize only the missing high-order channels. This design leverages the physical reliability of the observed data while focusing the generative capacity on resolving the underdetermined spatial details.

- **Extensive evaluation:** We conduct a comprehensive experimental study, including objective metrics and subjective listening tests across varied acoustic scenarios, demonstrating that the proposed generative framework improves the reconstruction of spatial details compared to existing discriminative baselines, particularly in challenging reverberant scenarios.

The rest of this paper is organized as follows. Section II introduces the signal model for Ambisonics and provides the mathematical foundation for continuous-time generative models. Section III details our proposed generative framework, describing the flexible hierarchical architecture, the targeted reconstruction strategy, and the signal preprocessing pipeline. The empirical validity of the approach is evaluated in Section IV, where we present a comprehensive numerical study across varied reverberant conditions followed by the results of our subjective listening tests in Sections V and VI. Finally, Section VII provides concluding remarks.

Throughout this paper, we denote multi-dimensional quantities using boldface notation, with their dimensions specified upon introduction. Calligraphic letters denote sets, e.g., $\mathcal{X}$, with $\mathbb{R}$ and $\mathbb{C}$ being the sets of real and complex numbers, while $\|\cdot\|$, $\|\cdot\|_F$, $(\cdot)^T$, $(\cdot)^H$, and $(\cdot)^\dagger$ are the ℓ_2 norm, Frobenius norm, transpose, conjugate transpose, and Moore-Penrose pseudo-inverse, respectively. For tensor indexing, $[\mathbf{X}]_{i_1, i_2, \dots, i_n}$ denotes a scalar entry of an n -dimensional tensor, while $[\mathbf{X}]_{:, \dots, :, i_k, :, \dots, :}$ denote sub-tensors obtained by fixing the k th index.

II. PRELIMINARIES AND PROBLEM FORMULATION

This section provides the background for AU. We begin by formulating the signal model and the fundamental properties of Ambisonics signals in Subsection II-A, followed by the formulation of the AU task as an inverse problem in Subsection II-B. Then, to motivate our methodology, we review the preliminaries of continuous-time generative models in Subsection II-C.

A. Signal Model and Ambisonics

Consider an array comprising M omni-directional microphones positioned at coordinates (r_i, θ_i, ϕ_i) for $1 \leq i \leq M$. The surrounding sound field is assumed to be composed of Q plane waves arriving from directions (θ_q, ϕ_q) , where $q \in \{1, \dots, Q\}$. The signal captured by the array at wave number k is expressed as

$$\mathbf{p}(k) = \mathbf{V}(k)\boldsymbol{\xi}(k) + \mathbf{n}(k), \quad (1)$$

where $\mathbf{p}(k) = [p_1(k), \dots, p_M(k)]^T$ is the $M \times 1$ vector of measured sound pressures, and $\mathbf{V}(k)$ is the $M \times Q$ array steering matrix whose (i, q) -th element represents the frequency response of the i -th microphone to a plane wave from direction q . The vector $\boldsymbol{\xi}(k) = [\xi_1(k), \dots, \xi_Q(k)]^T$ represents the Q source signals at the origin, and $\mathbf{n}(k)$ denotes additive i.i.d. microphone noise.

The Ambisonic representation of order N for the Q plane waves $\boldsymbol{\xi}(k)$ is given by [30]

$$\mathbf{a}_N(k) = \mathbf{Y}_Q^H \boldsymbol{\xi}(k), \quad (2)$$

where $\mathbf{a}_N(k)$ is a vector of size $(N+1)^2$ containing the Ambisonic components. The matrix $\mathbf{Y}_Q \in \mathbb{C}^{Q \times (N+1)^2}$ is comprised of the spherical harmonic functions sampled at the source directions (θ_q, ϕ_q) . To prevent spatial aliasing, the condition $k \cdot r \ll N$ must be satisfied, while $(N+1)^2 \leq M$ is required to facilitate Ambisonics encoding from microphone signals.

In practice, the plane wave signals $\xi(k)$ are not directly accessible, and the Ambisonic components are acquired from the measured $\mathbf{p}(k)$. In the case of a spherical microphone array, the steering matrix $\mathbf{V}(k)$ in (1) is decomposed as:

$$\mathbf{V}(k) = \mathbf{Y}_M \mathbf{B}(k) \mathbf{Y}_Q^H, \quad (3)$$

where $\mathbf{Y}_M \in \mathbb{C}^{M \times (N+1)^2}$ is the spherical harmonic matrix evaluated at the M microphone positions, and $\mathbf{B}(k)$ is a diagonal matrix of mode strength functions accounting for the array's radial characteristics. The Ambisonic signal of (2) is estimated by inverting the physical model, i.e., via

$$\hat{\mathbf{a}}_N(k) = (\mathbf{Y}_M \mathbf{B}(k))^{\dagger} \mathbf{p}(k). \quad (4)$$

While the representation in (4), termed *encoding*, is specific for spherical arrays, encoding techniques for general microphone arrays have been extensively studied, ranging from parametric spatial models to generalized neural encoders [31]–[33].

B. Problem Formulation

The Ambisonic order (N in (2)) dictates the spatial resolution of the sound field representation. While increasing N improves localization accuracy and reconstruction fidelity, it is physically constrained by the number of microphones M and the spatial aliasing limit. AU is defined as the task of reconstructing a HOA representation from a restricted low-order observation. Specifically, AU aims to map a low-order signal of order N into its high-order counterpart of order N' , where $N' > N$. This task can be mathematically framed as a linear inverse problem in the time-domain of the form

$$\mathbf{a}_N(\tau) = \mathbf{F} \mathbf{a}_{N'}(\tau), \quad (5)$$

where $\mathbf{a}_N(\tau) \in \mathbb{R}^{(N+1)^2}$ represents the captured time-domain low-order components and $\mathbf{a}_{N'}(\tau) \in \mathbb{R}^{(N'+1)^2}$ denotes the desired time-domain high-order components. Both Ambisonics signals are the inverse Fourier transform of the frequency-domain components defined in (2) over all wavenumbers k . The matrix $\mathbf{F} \in \mathbb{R}^{(N+1)^2 \times (N'+1)^2}$ is a truncation or selection matrix, typically structured as $\mathbf{F} = [\mathbf{I}_{(N+1)^2}, \mathbf{0}]$, which effectively discards the high-order spatial components.

Our aim in AU is to recover the high-order components from their low-order measurements observed via (5) over a time horizon $\tau_{\max}$, namely, for $\tau \in \{0, \dots, \tau_{\max}\}$. For design purposes, we assume access to a dataset comprised of multiple HOA recordings. The dataset is written as

$$\mathcal{D} = \left\{ \mathbf{A}_{N'}^{(i)} \right\}_{i=1}^{|\mathcal{D}|}. \quad (6)$$

where $\mathbf{A}_{N'}^{(i)} \in \mathbb{R}^{(N'+1)^2 \times \tau_{\max}}$ is the stacking of the time-domain high-order components $\{\mathbf{a}_{N'}^{(i)}(\tau)\}_{\tau=0}^{\tau_{\max}}$.

Because $\mathbf{F}$ is a wide matrix, the system in (5) is underdetermined, meaning there exists an infinite set of

high-order signals that satisfy the low-order observations. Traditionally, this is addressed using the least-norm solution via the Moore-Penrose pseudo-inverse [34]. However, as noted in [35], such deterministic solutions tend to distribute energy uniformly across the spatial domain, often leading to "spatial blurring," where the directional sharpness of the original plane waves is lost, ultimately degrading the localization cues in the rendered output. To recover the high-frequency spatial details lost during truncation, it is essential to incorporate prior knowledge regarding the structure of typical sound fields.

C. Preliminaries of Continuous-Time Generative Models

Continuous-time generative models are a dominant paradigm in generative learning [36]. The general framework of continuous-time generative models aims to learn a smooth transformation between a tractable prior distribution, p_0 (typically a simple Gaussian), and a complex target data distribution, p_1 , with both distributions defined over some vector space $\mathbb{R}^d$. This transition is conceptualized as a continuous path in probability space with time variable $t \in [0, 1]$, and specifically as a stochastic process $\{\mathbf{x}_t\}_{t \in [0, 1]}$ formulated via its differential $d\mathbf{x}_t$ such that $\mathbf{x}_0 \sim p_0$ and $\mathbf{x}_1 \sim p_1$. We next elaborate on two representative forms of such generative models: *FM* [29] and *SGMs* [37].

1) *Flow Matching*: In flow-based modeling, the transformation between distributions is defined by a transport map governed by an ordinary differential equation (ODE) of the form:

$$d\mathbf{x}_t = u_t(\mathbf{x}_t) dt, \quad t \in [0, 1], \quad (7)$$

where $u_t(\cdot)$ is a time-dependent vector field (VF) that ensures the marginals of $\mathbf{x}_t$ match the data distributions at $t = 0$ and $t = 1$. Since infinitely many paths can satisfy these boundary conditions, a common approach is to define a linear probability path

$$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1, \quad (8)$$

such that the conditional velocity field is expressed as

$$u_t(\mathbf{x}_t | \mathbf{x}_0, \mathbf{x}_1) = \frac{\mathbf{x}_1 - \mathbf{x}_0}{1 - t}. \quad (9)$$

The learning objective of FM is to train a DNN with trainable parameters θ , whose mapping $\mathbf{v}(\mathbf{x}_t, t; \theta)$ approximates this vector field. To avoid numerical instability caused by the singularity at $t = 1$ in (9), the model is typically trained to regress the constant displacement $(\mathbf{x}_1 - \mathbf{x}_0)$, which is derived by differentiating (8) with respect to t , via the Conditional Flow Matching (CFM) objective. This is shown in [29] to be equivalent in expectation to the FM objective.

Once the model $\mathbf{v}(\mathbf{x}_t, t; \theta)$ is trained, samples are generated by solving the ODE in (7) starting from a noise distribution $\mathbf{x}_0 \sim p_0$. Using the Euler method with J discrete time steps and a step size $\Delta t = 1/J$, the trajectory is computed as in Algorithm 1.

2) *Score-based Generative Modeling*: While FM relies on deterministic trajectories, SGM generalizes the continuous-time framework by framing the generative process as the solution to a reverse-time stochastic differential equation (SDE). In this setting, the transformation is not purely a velocity-driven

Algorithm 1: FM Sampling via Euler Method

Input : Number of steps J
 Prior distribution p_0
 Trained model parameters θ

```

1 Sample  $\mathbf{x}_0 \sim p_0$ ;
2 Set step size  $\Delta t \leftarrow 1/J$ ;
3 for  $j = 0$  to  $J - 1$  do
4   Set  $t_j \leftarrow j\Delta t$ ;
5    $\mathbf{u} \leftarrow \mathbf{v}(\mathbf{x}_{t_j}, t_j; \theta)$ ;
6    $\mathbf{x}_{t_{j+1}} \leftarrow \mathbf{x}_{t_j} + \mathbf{u}\Delta t$ ;
7 return  $\mathbf{x}_1$ 
```

transport map, but is augmented with a stochastic term based on Langevin dynamics. The reverse-time SDE is expressed as:

$$d\mathbf{x}_t = [u_t(\mathbf{x}_t, t) - g_t^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)] dt + g_t d\mathbf{w}_t, \quad (10)$$

where g_t is the diffusion coefficient, $d\mathbf{w}_t$ is a reverse-time Wiener process, and $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)$ is the score function of the marginal density at time t .

The learning objective is to train a DNN $\mathbf{s}(\mathbf{x}_t, t; \theta)$ to approximate the score function for all $t \in [0, 1]$, while treating $u_t(\cdot)$ as a predetermined function which depends on a specific diffusion family. This learning framework provides a unified perspective that encompasses several prominent model families of diffusion generative models, including Variance Exploding (VE), Variance Preserving, sub-Variance Preserving, and Ornstein-Uhlenbeck formulations, by appropriately configuring $u_t(\cdot)$ and g_t [38], [39]. Once trained, sampling is performed using Predictor-Corrector (P-C) methods [38, Appendix G], which combine numerical SDE solvers with Langevin-type MCMC updates as outlined in Algorithm 2. During the corrector step, the Langevin step size δ_j is adaptively determined at each iteration to maintain a constant signal-to-noise ratio.

Algorithm 2: SGM Predictor-Corrector Sampling

Input : Number of predictor steps J
 Prior distribution p_0
 Trained model parameters θ

```

1 Sample  $\mathbf{x}_0 \sim p_0$ ;
2 Set step size  $\Delta t \leftarrow 1/J$ ;
3 for  $j = 0$  to  $J - 1$  do
4   Set  $t_j \leftarrow j\Delta t$ ;
   // P Step: Euler-Maruyama SDE Solver
5    $\mathbf{z}_p \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ;
6    $\mathbf{f} \leftarrow (u_{t_j}(\mathbf{x}_{t_j}, t_j) - g_{t_j}^2 \mathbf{s}(\mathbf{x}_{t_j}, t_j; \theta))$ ;
7    $\mathbf{x} \leftarrow \mathbf{x}_{t_j} + \mathbf{f}\Delta t + g_{t_j} \sqrt{\Delta t} \mathbf{z}_p$ ;
   // C Step: Langevin MCMC
8    $\mathbf{z}_c \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ;
9    $\mathbf{x} \leftarrow \mathbf{x} + \delta_j \mathbf{s}(\mathbf{x}, t_{j+1}; \theta) + \sqrt{2\delta_j} \mathbf{z}_c$ ;
10  Set  $\mathbf{x}_{t_{j+1}} \leftarrow \mathbf{x}$ ;
11 return  $\mathbf{x}_1$ 
```

III. GENERATIVE AMBISONIC UPSCALING

In this section, we present our generative framework for AU. We begin in Subsection III-A by formulating the continuous-time generative models discussed in the previous section for spatial audio. Subsection III-B details our targeted reconstruction strategy, which focuses on completing missing high-order harmonics while preserving the physical integrity of the observation. The training pipeline is described in Subsection III-C, and is followed by a discussion in Subsection III-D.

A. Differential Equation Formulation for Ambisonics Upscaling

While the framework in Subsection II-C considers generic continuous-time generative modeling, we adapt it for the AU task. In our formulation, the sampled variable $\mathbf{x}_t$ represents the HOA components. We define a probability path that interpolates between a Gaussian noise prior $\mathbf{x}_0 \sim p_0$ and the target distribution $\mathbf{x}_1 \sim p_1$, which is the HOA distribution conditioned on the observed low-order components $\mathbf{y} = \mathbf{A}_N$. By governing the generative process via the conditional probability path $p_t(\mathbf{x}|\mathbf{y}) \triangleq p(\mathbf{x}_t = \mathbf{x}|\mathbf{y})$, the model reconstructs high-order spherical harmonics directly from low-order observations, effectively sampling from $p_1(\mathbf{x}|\mathbf{y})$. Consequently, the reconstruction of the missing Ambisonic orders based on the inverse problem (5) is constrained to physically plausible solutions through posterior sampling. This formulation specializes both the FM and SGM-oriented formulations for the differential equations governing the generative process, as detailed next:

1) *FM ODE*: In this framework, the VF component in the ODE (7) is conditioned on the observations. Accordingly, the underlying ODE is given by

$$d\mathbf{x}_t = u_t(\mathbf{x}_t|\mathbf{y})dt, \quad t \in [0, 1]. \quad (11)$$

Following the linear probability path defined in (8), the VF should regress the constant-velocity field required to map a sample from the Gaussian prior to a high-order completion $\mathbf{x}_1$, while being consistent with the low-order components $\mathbf{y}$.

2) *SGM SDE*: This setting requires a *conditional diffusion model* which must estimate the conditioned score function $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{y})$. The SDE we use is the VE SDE introduced in [38]. In this formulation, the VF is fixed to $u_t(\mathbf{x}_t, t) = 0$, while the diffusion term g_t is set to

$$g_t = \sigma_t \sqrt{2 \log \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)}, \quad (12a)$$

where σ_t is a geometric scheduler defined as:

$$\sigma_t = \sigma_{\min} \left(\frac{\sigma_{\max}}{\sigma_{\min}} \right)^t. \quad (12b)$$

Here, $\sigma_{\min}$ and $\sigma_{\max}$ are predefined noise level limits.

B. Proposed Generative Framework

1) *High Level Architecture*: The overall architecture of generative AU is illustrated in Fig. 1. While existing work on generative models for inverse problems typically focuses on spatial dimensions (in computer vision [40], [41]) or temporal and spectral dimensions (in audio [42], [43]), in AU the missing

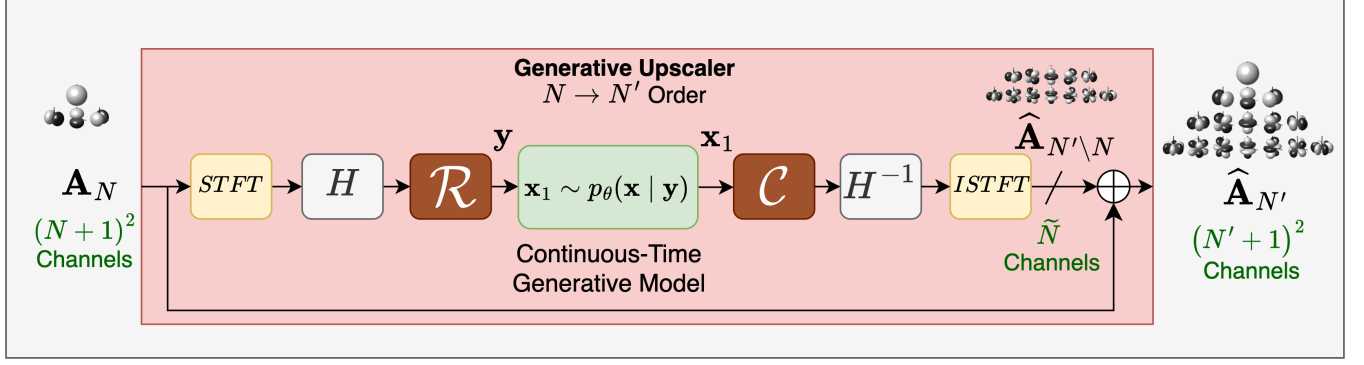


Fig. 1: Schematic illustration of the overall architecture of the proposed generative Ambisonics upscaler.

dimensions to be generated correspond to the restoration of the Ambisonics channel dimension. Specifically, we seek a *targeted reconstruction* of the channel dimension, i.e., rather than generating the full $(N' + 1)^2$ -channel signal from scratch, we allocate the entire generative capacity to the $\tilde{N} \triangleq (N' + 1)^2 - (N + 1)^2$ missing channels for orders $n \in \{N + 1, \dots, N'\}$. This formulation treats the AU task as a constrained inverse problem where the low-order components are fixed observations.

To solve this, we let $\hat{\mathbf{A}}_{N' \setminus N} \in \mathbb{R}^{\tilde{N} \times \tau_{\max}}$ denote the high-order components reconstructed by the model. The final estimate $\hat{\mathbf{A}}_{N'} \in \mathbb{R}^{(N' + 1)^2 \times \tau_{\max}}$ is then assembled by concatenating the physically measured low-order input with these generated channels as

$$\hat{\mathbf{A}}_{N'} = \begin{bmatrix} \mathbf{A}_N \\ \hat{\mathbf{A}}_{N' \setminus N} \end{bmatrix}. \quad (13)$$

This assembly strategy ensures that the low-order spatial information (where the physical measurement is most reliable) is fully preserved. Consequently, the generative model, which is guided by the measured low-order components $\mathbf{y}$ and follows the differential equations detailed in Subsection III-A, only needs to resolve the underdetermined portion of the inverse problem: the high-order spherical harmonics.

2) *Signal Representation*: The input is $\mathbf{A}_N$, a time-domain real Ambisonic signal of order N . We first transform the signal into the time-frequency (TF) domain via the short-time Fourier transform (STFT), and denote the resulting representation as $\mathbf{A}_N^{\text{tf}}$.

To address the heavy-tailed distribution of speech amplitudes and ensure consistent scaling for the DNN implementing the generative mapping, we apply a nonlinear amplitude transformation $\mathcal{H}(x)$ [39], [44], given by

$$\mathcal{H}(x) = \beta |x|^\alpha e^{j \arg(x)}, \quad \mathcal{H}^{-1}(x) = \frac{|x|^{1/\alpha}}{\beta} e^{j \arg(x)}, \quad (14)$$

where α and β are compression and scaling hyperparameters, respectively. Since the DNN operates in the real domain, we define a transformation $\mathcal{R}$ that concatenates the real and imaginary components along the channel dimension. Following the generative process, an inverse transformation $\mathcal{C}$ collapses these channels back into complex values, after which $\mathcal{H}^{-1}$ and the inverse STFT (ISTFT) are applied to recover the

time-domain HOA components. The complete AU procedure is summarized in Algorithm 3. While the underlying generative logic remains consistent with the frameworks described in Subsection II-C, the sampling process is now explicitly conditioned on the available low-order components $\mathbf{y}$.

Algorithm 3: Generative Upscaling

Input : $\mathbf{A}_N \in \mathbb{C}^{(N+1)^2 \times \tau_{\max}}$
Number of steps J
Generative model parameters θ

- 1 $\mathbf{A}_N^{\text{tf}} \leftarrow \text{STFT}(\mathbf{A}_N)$;
- 2 $\mathbf{y} \leftarrow \mathcal{R}\{\mathcal{H}(\mathbf{A}_N^{\text{tf}})\}$;
- 3 $\mathbf{x}_1 \leftarrow \text{Sampler}(\mathbf{y}, J, p_0, \theta)$ (via Algorithm 1 or 2) ;
- 4 $\hat{\mathbf{A}}_{N' \setminus N}^{\text{tf}} \leftarrow \mathcal{H}^{-1}(\mathcal{C}(\mathbf{x}_1))$;
- 5 $\hat{\mathbf{A}}_{N' \setminus N} \leftarrow \text{ISTFT}(\hat{\mathbf{A}}_{N' \setminus N}^{\text{tf}})$;
- 6 $\hat{\mathbf{A}}_{N'} \leftarrow \text{Concat}(\mathbf{A}_N, \hat{\mathbf{A}}_{N' \setminus N})$;
- 7 **return** $\hat{\mathbf{A}}_{N'} \in \mathbb{C}^{(N'+1)^2 \times \tau_{\max}}$

C. Training

The proposed training objective aims to recover the underlying conditional probability path $\{p_t\}_{t \in [0,1]}$, which provides the generative trajectory for the sampler in Step 3 of Algorithm 3. By approximating this path, the model ensures that the generated $\mathbf{x}_1$ constitutes a faithful recovery of the HOA components. Specifically, the learnable parameters θ are used to estimate the score function in SGMs or the VF in FM. In the following, we first describe the data preparation procedure, followed by the specific training losses for both frameworks.

1) *Data Preparation*: To this end, we sample a time-domain Ambisonic signal $\mathbf{A}_{N'}$ of order N' from the dataset $\mathcal{D}$ in (6) and partition it into its low- and high-order components. Specifically, we define the conditioning signal $\mathbf{y}$ using the first $(N + 1)^2$ channels, corresponding to the low-order signal $\mathbf{A}_N$. The remaining $\tilde{N}$ channels, representing the high-order components $\mathbf{A}_{N' \setminus N}$, are designated as the target data sample $\mathbf{x}_1$.

2) *Training Objective*: Due to the intractability of the continuous-time objective, these models are trained using Monte

Carlo estimation. Specifically, for each $(\mathbf{x}_1, \mathbf{y})$ pair, we draw a time step $t \sim \mathcal{U}(0, 1)$ and a noise sample $\mathbf{x}_0 \sim p_0$. We denote the collection of $|\mathcal{D}|$ sampled time steps and noise samples as $\mathbf{t}$ and $\mathbf{X}_0$, respectively. Based on these samples, the specific training objective varies by framework. The proposed training objectives for both FM and SGM operate on multiple channels, allowing the network to implicitly learn the underlying physical inter-channel dependencies. While one can in principle introduce additional regularization and explicitly encourage inter-channel correlation, we utilize the loss functions detailed in the sequel based on our empirical exploration and their established usefulness in training continuous-time generative models.

a) *Flow Matching*: In this framework, $\mathbf{x}_0$ which is sampled from $p_0 \equiv \mathcal{N}(\mathbf{0}, \mathbf{I})$, serves as the source of a generative path terminating at the data sample $\mathbf{x}_1$. To learn the corresponding VF, we adopt the *denoiser loss* formulation [45].

Specifically, the interpolated state $\mathbf{x}_t$ is constructed via (8), the model then predicts the target data sample $\mathbf{x}_1$ from the current state $\mathbf{x}_t$ through the estimated VF:

$$\hat{\mathbf{x}}_{1,t} = \mathbf{x}_t + (1 - t) \cdot \mathbf{v}(\mathbf{x}_t, t \mid \mathbf{y}; \boldsymbol{\theta}). \quad (15)$$

The resulting weighted empirical risk is given as:

$$\mathcal{L}_{\mathcal{D}, \mathbf{t}, \mathbf{X}_0}^{\text{CFM}}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{D}|N} \sum_{i=1}^{|\mathcal{D}|} w(t^{(i)}) \cdot \left\| \mathbf{x}_1^{(i)} - \hat{\mathbf{x}}_{1,t^{(i)}}^{(i)} \right\|^2. \quad (16)$$

While a weighting of $w(t) = (1 - t)^{-2}$ recovers the original velocity-regression objective of [29], we employ a logarithmic scheme $w(t) = \log(1 + t)$, inspired by practices employed in training deep unfolded models [46]. Empirically, this was found to enhance training stability and generative quality.

b) *Score-based Generative Model*: This approach adopts the *denoising score matching* strategy [47] to learn the score of the conditional marginal density, i.e., $\nabla p_t(\mathbf{x} \mid \mathbf{y})$. The interpolated state is parameterized with the VE perturbation kernel:

$$\mathbf{x}_t = \mathbf{x}_1 + \sigma_t \mathbf{x}_0, \quad (17)$$

where the scheduler σ_t , as defined in (12b), interpolates between the data distribution at $t = 1$ and the prior at $t = 0$, which is $p_0 \equiv \mathcal{N}(\mathbf{0}, \sigma_{\max}^2 \mathbf{I})$ [48]. The empirical loss is defined as:

$$\mathcal{L}_{\mathcal{D}, \mathbf{t}, \mathbf{X}_0}^{\text{SGM}}(\boldsymbol{\theta}) = \frac{1}{|\mathcal{D}|N} \sum_{i=1}^{|\mathcal{D}|} \left\| \mathbf{s}(\mathbf{x}_{t^{(i)}}^{(i)}, t^{(i)} \mid \mathbf{y}^{(i)}; \boldsymbol{\theta}) \cdot \sigma_{t^{(i)}} + \mathbf{x}_0^{(i)} \right\|^2. \quad (18)$$

3) *Training Procedure*: The overall training procedure based on mini-batch stochastic gradient descent (SGD) is detailed in Algorithm 4. The formulation of Algorithm 4 accommodates both FM and SGM for generative modeling. There, we also maintain a shadow copy of the model weights, used for inference, which is updated via Exponential Moving Average (EMA) with forgetting coefficient $\gamma \in [0, 1]$ to improve generalization and robustness.

D. Discussion

The proposed methodology introduces a novel approach to AU that is particularly well-suited to the formulation of the task in Subsection II-B, which is inherently underdetermined. Unlike deterministic approaches that seek a single point estimate, often

Algorithm 4: Training Generative AU via SGD

Init: Model parameters $\boldsymbol{\theta}$; $\boldsymbol{\theta}_{\text{EMA}} = \boldsymbol{\theta}$
 Learning rate η ; EMA weight γ
Input: Training set $\mathcal{D} = \{\mathbf{A}_{N'}^{(i)}\}_{i=1}^{|\mathcal{D}|}$
 1 Partition data into high- and low-order $\mathcal{D} = \{\mathbf{x}_1^{(i)}, \mathbf{y}^{(i)}\}$;
 2 **for** epoch = 0, 1, ... **do**
 3 Divide $\mathcal{D}$ into B batches $\{\mathcal{D}_b\}_{b=1}^B$;
 4 **for** $b = 1, \dots, B$ **do**
 5 Draw $|\mathcal{D}_b|$ i.i.d. time steps $\sim \mathcal{U}(0, 1)$ as $\mathbf{t}$;
 6 Draw $|\mathcal{D}_b|$ i.i.d. samples $\sim p_0$ as $\mathbf{X}_0$;
 7 Compute loss $\mathcal{L}_{\mathcal{D}_b, \mathbf{t}, \mathbf{X}_0}(\boldsymbol{\theta})$ via (16) or (18);
 8 Set $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \nabla \mathcal{L}_{\mathcal{D}_b, \mathbf{t}, \mathbf{X}_0}(\boldsymbol{\theta})$;
 9 Set $\boldsymbol{\theta}_{\text{EMA}} \leftarrow \gamma \boldsymbol{\theta} + (1 - \gamma) \boldsymbol{\theta}_{\text{EMA}}$
 10 **return** $\boldsymbol{\theta}_{\text{EMA}}$

leading to spatial blurring, the formulation in Subsection III-B recasts AU as conditional generative inference. By modeling the conditional distribution of the missing high-order components given the observed low-order signal, the framework explicitly captures the ambiguity and stochasticity induced by reverberation. The targeted reconstruction strategy further aligns with the physical structure of Ambisonics by preserving the reliably measured low-order components while restricting the generative process to the missing channels. As a result, the generated HOA signals remain consistent with both the observed sound field and the underlying physical constraints, while enabling the synthesis of spatial features that reflect the true data distribution. The empirical results presented in Section IV corroborate this design, demonstrating that the proposed approach achieves accurate and robust upscaling across a wide range of acoustic conditions, including highly reverberant environments.

Beyond the specific instantiations considered in this work, the proposed formulation naturally admits several extensions. While we focus on instances of FM and Diffusion-based models, the framework is compatible with a broad class of continuous-time generative approaches. Moreover, the current formulation assumes static acoustic scenes; extending it to account for temporal dynamics, such as moving sources or listener motion, constitutes a promising direction. Another avenue is to move beyond signal reconstruction objectives and directly optimize for downstream perceptual tasks, such as binaural rendering, thereby aligning the generative process with end-use criteria. Finally, although the results indicate that a single trained model generalizes well across related acoustic conditions, further improvements may be achieved by enabling adaptation to diverse environments, for example via online updating mechanisms or hypernetwork-based conditioning. We leave these directions for future work.

IV. EMPIRICAL EVALUATION

In this section, we evaluate the performance of the proposed continuous-time generative models for AU¹. The objective is to upscale a FOA signal ($N = 1$) to a 3rd-order representation ($N' = 3$) denoted as HOA3. The evaluation is structured as

¹The source code is available at <https://github.com/Amitmils/GenAU>.

follows: Subsection IV-A outlines the experimental setup and the datasets used for training and evaluation. Subsection IV-B details the methodology, including the model configurations, the AU baselines, and the evaluation metrics. Finally, Subsection IV-C reports the results of our numerical analysis.

A. Experimental Setup

1) *Training Data*: We utilize a speech dataset comprising 100 hours of audio, where each sample is 2 s in length and sampled at 16 kHz. In total, we employed 7,600 Spherical Harmonics Room Impulse Response (SH-RIR) derived from two distinct datasets:

- **MOTUS** [49]: A collection of real-world recordings comprising over 800 room configurations within a $4.9 \text{ m} \times 4.4 \text{ m} \times 2.9 \text{ m}$ space. For each configuration, four SH-RIRs were recorded from fixed source directions. To ensure a vast diversity of acoustic scenes, the internal geometry was randomized by repositioning furniture and altering wall and floor materials, resulting in Reverberation Time (T_{60}) primarily concentrated between 500 ms and 1.2 s.
- **HARP** [50]: A dataset of simulated SH-RIRs with T_{60} concentrated between 200 ms and 1.5 s. These SH-RIRs were generated in shoebox geometries via the image source method (reflection order 30) using the HARP codebase which utilizes the `pyroomacoustics` library [51]. The dataset encompasses 39 rooms with volumes ranging from small (30 m^3) to large (850 m^3). While the source and receiver were maintained at a constant 1 m distance, azimuth and elevation angles were randomized. To ensure acoustic consistency, the source was placed at least 0.5 m from any wall, and the receiver position remained fixed for all simulations within a specific room configuration.

For each room configuration, we randomly selected between one and four SH-RIRs and convolved them with dry audio samples from the WSJ0 corpus [52]. To enhance model robustness, a random gain between -6 dB and 6 dB was applied to each source. Furthermore, during training, we performed random rotations on the sound field to ensure the model achieves rotation invariance. All signals were processed in 3rd-order Ambisonics following the N3D [4] normalization convention.

2) *Evaluation Data*: We evaluate the proposed models on unseen room configurations from the HARP and MOTUS datasets, ensuring zero overlap between training and evaluation environments. To facilitate a comprehensive performance analysis, we simulate with HARP codebase a wide range of acoustic scenarios with T_{60} values from 0.1 s to 1.25 s and Direct-to-Reverberant-Ratio (DRR) levels from -5 dB to 10 dB . This 2-hour recording set, whose distribution is shown in Fig. 2, is partitioned into subsets to analyze performance across different physical metrics. Specifically, the T_{60} subsets are categorized as *Low* ($T_{60} < 300 \text{ ms}$), *Medium* ($300 \text{ ms} \leq T_{60} < 700 \text{ ms}$), and *High* ($T_{60} \geq 700 \text{ ms}$). Similarly, the DRR subsets are divided into *Low* ($\text{DRR} < 0 \text{ dB}$), *Medium* ($0 \text{ dB} \leq \text{DRR} < 5 \text{ dB}$), and *High* ($\text{DRR} \geq 5 \text{ dB}$).

Complementing this broad analysis, we also evaluate on the MOTUS dataset. This set provides a more focused benchmark for a constrained range of acoustic conditions, specifically featuring rooms with T_{60} between 0.5 s and 1.0 s and DRR

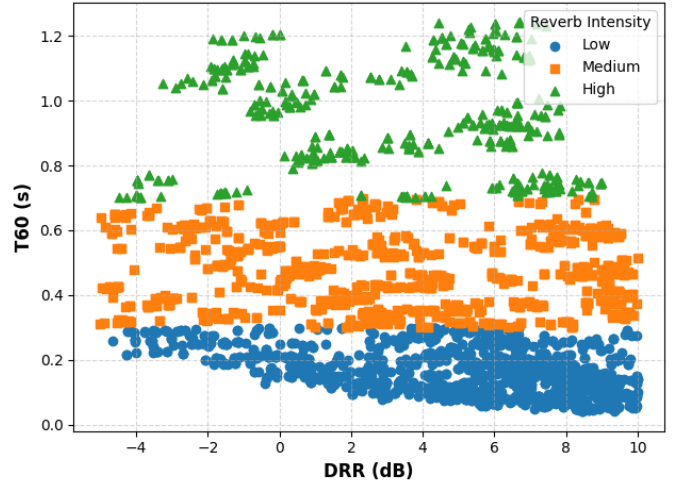


Fig. 2: HARP evaluation data distribution

values between -5 dB and 0 dB . Furthermore, to evaluate robustness to the number of sources, we partitioned our datasets by the number of speakers, which ranges from 1 to 4.

As model-based AU approaches assume free-field scenarios, we also consider anechoic settings. There, we randomly selected 1 – 4 speakers for each signal, and for each selected speaker, a random directions of arrival (DOA) was assigned, and the corresponding HOA was constructed via (2). It is emphasized that we do not retrain the data-driven algorithms differently in anechoic settings, and use the same models trained using the reverberant dataset detailed in Subsection IV-A1 for all scenarios.

B. Methodology

We implement our generative framework utilizing the noise-conditioned score-matching network (NCSN++) backbone [38] with modifications to support multi-channel inputs and outputs specifically tailored to the AU configuration as detailed in Subsection III-B. To accommodate the high capacity required by generative frameworks to accurately model the complex data distribution of multi-channel sound fields, we employ a hidden dimension of 256, resulting in 262M parameters. While this represents a higher parameter footprint than the more compact, deterministic baselines evaluated, this capacity is essential for capturing the intricate spatial and phase relationships necessary for high-fidelity generation of complex reverberant scenarios. This unified backbone is employed for both the SGM (where it implements the score DNN $s(\cdot)$), termed *Diffusion*, and FM (where it estimates the VF $v(\cdot)$), termed *Flow Matching*. For the signal representation, we employ $\alpha = 0.5$ and $\beta = 0.15$ to compress and scale the signal before and after the network, respectively. We use $J = 3$ step in the FM sampler (Algorithm 1), resulting in 3 function evaluations, and $J = 25$ steps for the SGM sampler (Algorithm 2), which requires 50 function evaluations.

1) *Benchmarks*: We compare our generative framework to leading data-driven and model-based AU algorithms:

- **CTN**: Based on the discriminative model proposed in [24]. We utilize a configuration with 384 encoder channels and

a single repetition of 256 channels which totals in 3.5M parameters. The model accepts FOA signals as input and targets the reconstruction of missing HOA3 channels ($\tilde{N} = 12$), operating in a manner analogous to our proposed framework.

- *Cascaded GRU*: The model proposed in [23] utilizes cascaded GRUs corresponding to each Ambisonic order. Each stage consists of a single GRU unit followed by a linear layer that projects the output to the required number of channels for that specific order. Following the original architecture, the hidden dimension is set to 128. We evaluate two variants: the unidirectional version as originally proposed, and a bidirectional implementation, which grants the model a non-causal advantage consistent with the other methods under test. We shall denote them as GRU-Unidirectional (GRU-Uni) and GRU-Bidirectional (GRU-Bi), which in total have 0.2M and 0.4M parameters, respectively.
- *PWD CS*: The model-based framework proposed in [10] which casts AU as CS. This approach minimizes the $L_{1,2}$ -norm of the signals in the plane-wave domain to promote spatial sparsity. The reconstruction is subject to FOA consistency constraints, which are incorporated via Lagrange multipliers and solved using gradient descent.
- *HO-DirAC*: A parametric method proposed in [13] that assumes a single source per TF tile, estimating one DOA and diffuseness from FOA signals to steer HOA covariance-matched resynthesis.
- *COMPASS*: A parametric method proposed in [14] that accommodates multiple simultaneous sources per TF tile, estimating multiple DOAs and diffuseness parameters to reconstruct the target HOA components via adaptive mixing.

To ensure a fair comparison, all data-driven models were trained to convergence on the same dataset.

2) *Metrics*: To assess signal reconstruction, we use the following metrics on the generated channels:

- *STFT signal-to-distortion ratio (STFT-SDR)*, defined as

$$\text{STFT-SDR}(\hat{\mathbf{A}}_{3\setminus 1}) = 10 \cdot \log_{10} \left(\frac{\|\mathbf{A}_{3\setminus 1}^{\text{tf}}\|_F^2}{\|\mathbf{A}_{3\setminus 1}^{\text{tf}} - \hat{\mathbf{A}}_{3\setminus 1}^{\text{tf}}\|_F^2} \right), \quad (19)$$

where $\mathbf{A}_{3\setminus 1}^{\text{tf}}$ and $\hat{\mathbf{A}}_{3\setminus 1}^{\text{tf}}$ represent the ground truth and estimated high-order components (2nd and 3rd order) of the Ambisonic signals in the TF domain. A higher STFT-SDR value indicates better signal reconstruction.

- *Upscaling coherence*: to evaluate the reconstruction quality across the frequency spectrum, we calculate the magnitude-squared coherence between the upscaled signal and the ground truth, averaged across the HOA components. The coherence at frequency f is defined as:

$$\bar{\gamma}_f^2(\hat{\mathbf{A}}_{3\setminus 1}) = \frac{1}{12} \sum_{n=0}^{11} \frac{\left| [\hat{\mathbf{A}}_{3\setminus 1}^{\text{tf}}]_{n,:f}^H [\mathbf{A}_{3\setminus 1}^{\text{tf}}]_{n,:f} \right|^2}{\left\| [\mathbf{A}_{3\setminus 1}^{\text{tf}}]_{n,:f} \right\|^2 \cdot \left\| [\hat{\mathbf{A}}_{3\setminus 1}^{\text{tf}}]_{n,:f} \right\|^2}. \quad (20)$$

This metric evaluates spectral fidelity as a function of frequency, where a value of 1 represents full reconstruction.

While (19)-(20) measure signal reconstruction fidelity, they do not fully characterize the spatial resolution of the upscaled sound

field. To this end, we investigate the spatial accuracy of the up-scaling in single speaker scenarios using two key spatial metrics:

- *Directivity Index (DI) error*: The DI [53] quantifies the ability to extract a directional source from a diffuse field. For an order- N Ambisonic signal $\mathbf{A}_N$, the DI is

$$\text{DI}(\mathbf{A}_N) = 10 \cdot \log_{10} \left(\frac{\max_{\Omega} \|\mathbf{v}(\Omega)^T \mathbf{A}_N\|^2}{\frac{1}{4\pi} \|\mathbf{A}_N\|_F^2} \right) \quad (21)$$

where $\mathbf{v}(\Omega) \in \mathbb{R}^{(N+1)^2}$ is the N3D steering vector consisting of real spherical harmonics up to order N , for direction $\Omega = (\theta, \phi)$, with θ and ϕ denoting the colatitude and azimuth, respectively. The DI error is computed as

$$\Delta \text{DI}(\hat{\mathbf{A}}_N) = |\text{DI}(\hat{\mathbf{A}}_N) - \text{DI}(\mathbf{A}_N)|. \quad (22)$$

This metric penalizes both spatial under-estimation (blurring) and over-estimation (artificial de-reverberation), where a lower value indicates better spatial reconstruction fidelity.

- *Angular difference ($\Delta\sigma$)*, defined as the great-circle distance between the DOA of the estimated and ground-truth signals. The DOA is estimated by finding the maximum intensity on a dense spherical grid:

$$(\hat{\theta}, \hat{\phi}) = \arg \max_{\Omega} \|\mathbf{v}(\Omega)^T \mathbf{A}_N\|^2, \quad (23)$$

with which the angular error is then calculated as [4]:

$$\Delta\sigma = \cos^{-1} \left(\sin \theta \sin \hat{\theta} + \cos \theta \cos \hat{\theta} \cos(\phi - \hat{\phi}) \right), \quad (24)$$

where (θ, ϕ) are the ground-truth DOA derived via an identical grid search performed on the reference HOA3 signal.

While ΔDI characterizes the reconstructed spatial resolution, it does not account for the directional accuracy of the soundfield. Thus, by evaluating both ΔDI and $\Delta\sigma$, we provide a holistic assessment of the model's ability to not only sharpen the spatial response but also accurately preserve the source's position.

C. Results

1) *Overall Performance Under Anechoic Conditions*: While our methodology primarily targets complex reverberant environments, we begin with an anechoic signal reconstruction analysis to evaluate the frameworks where model-based assumptions are expected to hold. Table I shows STFT-SDR for the methods under study with one to four speakers. As shown in the table, the parametric methods (HO-DirAC and COMPASS) achieve the highest scores for a single speaker, as the anechoic single-source condition strictly satisfies the single-dominant-source assumption of HO-DirAC while providing an ideal low-complexity environment for COMPASS. However, their performance drops sharply with more speakers due to overlapping spatial components increasing the complexity for parameter estimation. Similarly, the spatial sparsity assumptions of PWD CS hold reasonably well across all mixtures, maintaining stable performance. Conversely, data-driven models show robustness to anechoic multi-speaker interference despite receiving no explicit anechoic training. Notably, in these low-uncertainty scenarios with 1-2 speakers, the diffusion framework underperforms the other data driven baselines. We attribute this to the low spatial

TABLE I: STFT-SDR (dB) evaluated for all methods under anechoic conditions, for 1-4 speakers. Best results are bolded.

Model	1 Spk	2 Spk	3 Spk	4 Spk	Avg.
PWD CS	12.90	14.34	12.38	10.94	12.63
HO-DirAC	37.87	13.97	7.89	5.22	15.65
COMPASS	40.57	15.95	9.93	7.24	17.82
GRU-Uni	11.50	9.04	7.06	5.63	8.31
GRU-Bi	12.30	9.89	7.89	6.40	9.12
CTN	14.05	11.77	9.76	8.60	10.97
Diffusion	10.74	10.59	9.75	9.09	10.02
Flow Matching	14.96	16.02	15.14	14.02	15.03

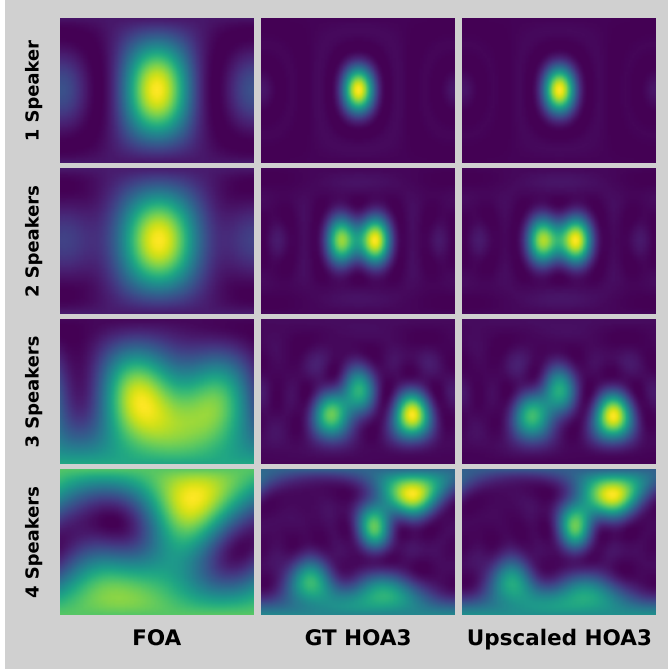


Fig. 3: Directional energy plots over elevation (x-axis) and azimuth (y-axis). Columns: FOA, ground truth HOA3 and FM upscaled HOA3. Rows correspond to the number of active sources.

ambiguity of the anechoic setting; under these conditions, a direct conditional mean-regression approach is highly optimal. In contrast, because the diffusion model employs an SDE sampling process designed to escape global averages, the continuous injection of stochastic noise introduces random structural fluctuations. This stochastic variance degrades both the fine spectral magnitude features and phase alignment, causing it to underperform. Reaffirming this, the FM framework, which instead resolves the generative process via a deterministic ODE trajectory, retains a highly structured path that tracks the conditional mean velocity field without random spectral or phase degradations.

The anechoic setting also facilitates clear visualization of spatial directivity. Accordingly, we depict in Fig. 3 the directional energy plots of 3rd-order Ambisonics signals produced by FM upscaler with the ground truth for orders 1 and 3, across one to four active speakers. The results show a strong resemblance between the recovered HOA and the reference energy patterns.

2) *Overall Performance Under Reverberant Conditions:* We now proceed to the primary focus of our experimental study: the evaluation of performance within complex reverberant

environments. A high-level summary, representing the results averaged across all T_{60} and DRR subsets of the HARP dataset, is provided in Table II.

TABLE II: STFT-SDR (dB) across T_{60} and DRR levels for the HARP evaluation set, detailed for 1–4 Speakers.

Model	1 Spk	2 Spk	3 Spk	4 Spk	Avg.
PWD CS	2.34	2.06	1.85	1.70	1.99
HO-DirAC	3.09	1.45	0.71	0.21	1.36
COMPASS	5.28	3.35	2.52	1.97	3.27
GRU-Uni	5.46	3.82	3.03	2.60	3.73
GRU-Bi	5.71	4.22	3.43	2.97	4.08
CTN	6.37	5.16	4.31	3.74	4.90
Diffusion	6.26	4.98	4.04	3.49	4.69
Flow Matching	8.79	7.97	7.08	6.26	7.53

Performance across all methods degrades in reverberant scenarios relative to anechoic conditions, which is expected given the highly stochastic nature of reverberant data. Specifically, the model-based methods exhibit significant degradation, as their underlying signal-model assumptions no longer hold in these complex acoustic environments. Consequently, the remainder of our evaluation focuses primarily on data-driven methods.

The results indicate that the Diffusion framework achieves performance comparable to the CTN baseline, both of which are significantly outperformed by the FM framework across all speaker conditions. We attribute this performance gap to the fundamental differences in how these two generative frameworks model and navigate the data manifold. Standard signal reconstruction metrics, such as STFT-SDR, operate on point-wise structural alignment and heavily penalize any phase or realization mismatch. Because the deterministic ODE formulation of FM tends to seek the conditional mean, it effectively synthesizes a smooth, expected representation of the target signal, maximizing average reconstruction metrics like STFT-SDR. In contrast, the diffusion model generates distinct, stochastic realizations of the true posterior. While this preserves natural acoustic textures, the resulting per-realization variance manifests as a penalty in strict point-wise distortion metrics, artificially widening the observed performance gap. Crucially, as we demonstrate in subsequent evaluations, this gap narrows significantly when shifting to spatial and perceptual criteria.

A similar trend is observed in the overall performance on the MOTUS dataset presented in Table III, characterized by a more substantial performance gap between the CTN baseline and the diffusion framework, while the FM framework consistently achieves the best performance.

TABLE III: STFT-SDR (dB) across T_{60} and DRR levels for the MOTUS evaluation set, detailed for 1–4 Speakers.

Model	1 Spk	2 Spk	3 Spk	4 Spk	Avg.
GRU-Uni	1.22	1.26	1.28	1.15	1.23
GRU-Bi	2.09	2.15	2.21	2.02	2.12
CTN	3.47	3.25	3.12	2.77	3.15
Diffusion	2.66	2.30	2.09	1.63	2.17
Flow Matching	5.98	5.56	5.32	4.70	5.39

3) *The Effect of T_{60} and DRR on Performance:* We now present a comprehensive analysis using the HARP dataset, which encompasses a wide range of reverberant scenarios to

TABLE IV: STFT-SDR (dB) for the HARP evaluation set, divided by Reverberation (Rev.) and DRR levels for 1–4 speakers. Best results are bolded.

Rev.	DRR	Model	1 Spk	2 Spk	3 Spk	4 Spk	Avg.
Low	Low	GRU-Uni	2.16	1.69	1.27	1.20	1.61
		GRU-Bi	2.26	1.86	1.39	1.33	1.74
		CTN	2.53	2.15	1.68	1.55	1.98
		Diffusion	2.47	2.03	1.39	1.06	1.74
		Flow Matching	4.29	3.96	3.31	2.96	3.63
	Med	GRU-Uni	4.48	3.33	2.85	2.51	3.30
		GRU-Bi	4.63	3.62	3.16	2.82	3.56
		CTN	5.07	4.24	3.75	3.36	4.11
		Diffusion	5.55	4.63	3.95	3.46	4.40
		Flow Matching	7.15	6.55	5.87	5.34	6.23
	High	GRU-Uni	7.50	5.10	4.01	3.37	5.00
		GRU-Bi	7.77	5.67	4.60	3.93	5.49
		CTN	8.73	7.03	5.95	5.07	6.70
		Diffusion	8.72	7.15	6.11	5.20	6.80
		Flow Matching	11.53	10.23	9.02	7.99	9.69
	Low	GRU-Uni	2.10	1.69	1.32	1.14	1.54
		GRU-Bi	2.24	1.87	1.50	1.31	1.71
		CTN	2.53	2.20	1.84	1.56	2.03
		Diffusion	2.35	2.00	1.47	1.10	1.73
		Flow Matching	3.45	3.35	2.87	2.61	3.07
	Med	GRU-Uni	4.25	3.10	2.63	2.25	3.07
		GRU-Bi	4.45	3.46	3.00	2.60	3.39
		CTN	5.20	4.32	3.74	3.27	4.13
		Diffusion	5.61	4.69	4.05	3.37	4.43
		Flow Matching	7.31	6.64	6.06	5.23	6.31
	High	GRU-Uni	7.82	5.26	4.04	3.74	5.22
		GRU-Bi	8.10	5.89	4.61	4.30	5.73
		CTN	9.58	7.71	6.28	5.90	7.37
		Diffusion	9.00	7.56	6.47	6.07	7.28
		Flow Matching	11.77	10.59	9.38	8.92	10.17
High	Low	GRU-Uni	2.40	1.73	1.52	1.33	1.69
		GRU-Bi	2.59	1.93	1.72	1.53	1.89
		CTN	3.13	2.54	2.16	1.93	2.44
		Diffusion	3.02	2.29	1.86	1.47	2.16
		Flow Matching	4.89	4.10	3.70	3.34	4.01
	Med	GRU-Uni	4.17	3.06	2.49	2.08	2.97
		GRU-Bi	4.41	3.44	2.82	2.41	3.29
		CTN	5.18	4.30	3.55	3.18	4.05
		Diffusion	5.17	4.39	3.48	3.11	4.04
		Flow Matching	7.28	6.49	5.77	5.24	6.20
	High	GRU-Uni	6.88	4.62	3.88	3.08	4.63
		GRU-Bi	7.17	5.21	4.45	3.57	5.10
		CTN	8.25	6.79	5.88	4.89	6.45
		Diffusion	7.35	6.49	5.72	4.94	6.13
		Flow Matching	9.53	8.85	8.13	7.29	8.45

more rigorously evaluate model performance across diverse acoustic conditions. Table IV details the performance of the various AU methods across the specific acoustic categories defined in Subsection IV-A2.

The results indicate that DRR exerts a significantly more profound influence on performance than either T_{60} or the number of speakers. To further illustrate this behavior, Fig. 4 depicts performance across a continuous range of DRR values, grouped by the defined T_{60} subsets for single-speaker scenarios. Notably, the performance curves for the various T_{60} subsets exhibit nearly identical trajectories; this reinforces the premise that reconstruction quality is primarily driven by DRR, maintaining a consistent trend across model architectures regardless of the specific reverberation category. These trends indicate that while the complexity of the acoustic scene increases with more speakers, the primary bottleneck for accurate signal reconstruction is the DRR energy rather than the temporal decay characteristics of the room.

TABLE V: Spatial performance in terms of ΔDI and $\Delta\sigma$ across DRR levels. Lower values indicate better reconstruction of the spatial field. Best results are bolded.

DRR	Model	ΔDI (dB) ↓	$\Delta\sigma$ (°) ↓
Low	Input (FOA)	5.25	4.68
	GRU-Uni (HOA-3)	1.16	3.33
	GRU-Bi (HOA-3)	1.23	3.14
	CTN (HOA-3)	1.46	2.61
	Diffusion (HOA-3)	0.23	1.51
	Flow Matching (HOA-3)	0.90	0.45
Med	Input (FOA)	5.62	2.89
	GRU-Uni (HOA-3)	0.71	2.16
	GRU-Bi (HOA-3)	0.74	1.96
	CTN (HOA-3)	0.76	1.68
	Diffusion (HOA-3)	0.11	1.22
	Flow Matching (HOA-3)	0.40	0.83
High	Input (FOA)	5.86	1.85
	GRU-Uni (HOA-3)	0.28	1.52
	GRU-Bi (HOA-3)	0.30	1.36
	CTN (HOA-3)	0.19	1.11
	Diffusion (HOA-3)	0.06	1.06
	Flow Matching (HOA-3)	0.14	0.67

4) *Frequency-Dependent Coherence*: We proceed to evaluate the performance across the frequency spectrum for all methods, with the coherence averaged over channels, as defined in (20), which is illustrated in Fig. 5. Notably, the generative approaches maintain consistent performance across the entire spectrum, whereas the discriminative ones degrade significantly at higher frequencies. This decay likely stems from the use of deterministic regression losses, which tend to average out high-frequency phase details to minimize point-wise error under uncertainty. In contrast, generative models leverage learned data distributions to synthesize realistic high-frequency components, maintaining structural coherence.

5) *Spatial Performance Analysis*: Lastly, we evaluate spatial performance of the AU methods. We evaluate the spatial accuracy, prioritizing a per-DRR analysis to highlight the governing factors of the spatial reconstruction.

Table V presents the spatial performance of each approach relative to the ground-truth HOA3 on reverberant single-speaker scenarios, with FOA performance included as a reference baseline. As illustrated, the generative approaches exhibit spatial characteristics that more closely align with the ground truth than those of the baseline models. We attribute this advantage to the fundamental differences in their optimization objectives. While the discriminative baseline minimizes a global reconstruction error, it inherently suffers from regression to the mean, resulting in spatially blurred outputs where fine-grained directional information is lost. In contrast, by modeling the score function or the probability flow of the HOA data, the generative frameworks successfully reconstruct signals that adhere to the true data manifold, thereby preserving delicate spatial characteristics.

Moreover, a distinct trade-off between the two generative paradigms emerges in Table V: while the SGM framework achieves a lower ΔDI compared to FM, the trend for $\Delta\sigma$ is reversed. This phenomenon stems from a subtle distinction in how these two methods sample the generative space. Unlike the global mean-regression of the discriminative baseline, the deterministic ODE trajectories of FM track the local conditional mean when confronted with intersecting generation paths. This

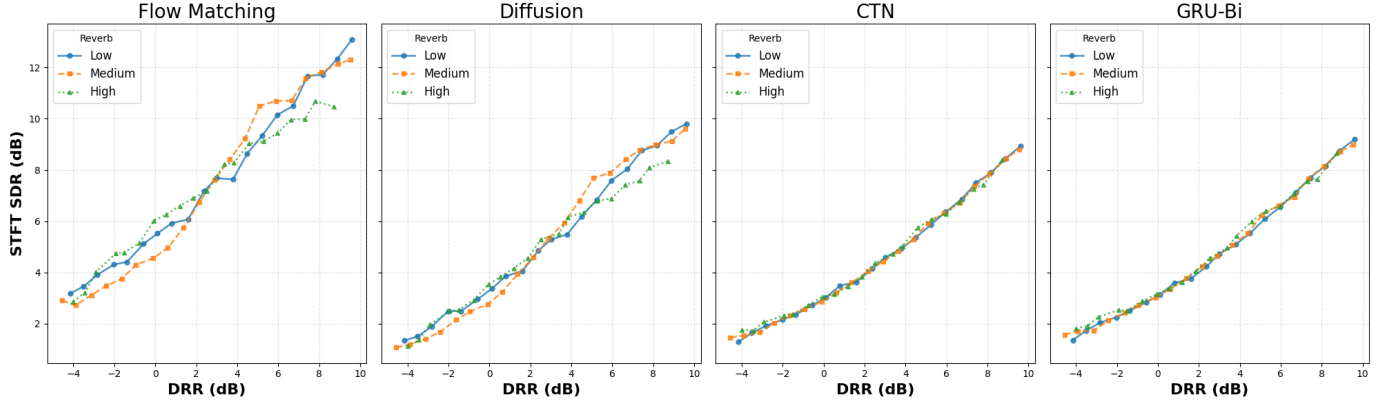


Fig. 4: Comparative analysis of STFT-SDR (dB) performance across models as a function of DRR, categorized by T_{60} levels.

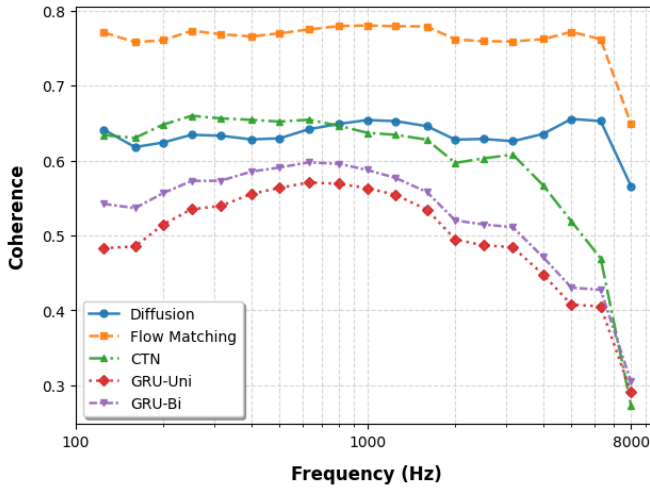


Fig. 5: Average coherence $\bar{\gamma}_f^2$ versus frequency.

condition becomes increasingly prevalent in highly reverberant data, introducing localized spatial blurring that manifests as an increased ΔDI . By comparison, SGM acts as a stochastic, unbiased sampler during inference; its SDE formulation continuously injects random noise, enabling the generative process to “escape” these local mean-seeking intersections and converge toward a single, sharp realization of the true posterior. This interpretation is further validated by the fact that the gap in ΔDI widens as the DRR decreases, suggesting a higher density of trajectory intersections in heavily reverberant environments for FM path.

V. LISTENING TEST I - ACOUSTIC SCENARIO COMPARISON

To complement the previous quantitative evaluations, listening tests were conducted². While metrics reported in Section IV are essential for quantitatively assessing AU in terms of signal reconstruction and directivity, they do not always capture the full complexity of human auditory perception. This study serves as an investigation into the subjective quality of our proposed method, exploring whether the observed technical trends across acoustic scenarios are reflected in listener preferences. To that

aim, we first describe the listening test setup and methodology in Subsections V-A and V-B, respectively, while Subsection V-C provides the results and a discussion of the findings.

A. Setup

All signals were generated by convolving 6-second dry speech samples from the WSJ0 corpus with simulated SH-RIRs from the HARP codebase. Notably, both the dry signals and the specific room configurations used for the SH-RIR simulations were withheld from the training phase to ensure data integrity.

The evaluation comprised two distinct listening tests, totaling seven acoustic scenarios. This design was implemented to determine if subjective perception aligns with the quantitative trends observed in Subsection IV-C. To isolate the acoustic environment as the primary variable, the speaker identity, utterance, and source position were kept constant within each test group:

- 1) **DRR Variation Group:** Four scenarios with a fixed T_{60} of 500 ms were evaluated to determine the impact of DRR variation across different room scales. A single male speaker was selected, positioned at azimuth of -60° and an elevation of 90° :
 - Scenario A:** $6.8 \times 5.0 \times 2.5$ m (-2.5 dB DRR)
 - Scenario B:** $6.2 \times 8.0 \times 2.6$ m (0.0 dB DRR)
 - Scenario C:** $12.0 \times 9.0 \times 2.2$ m (3.5 dB DRR)
 - Scenario D:** $17.6 \times 21.2 \times 4.1$ m (9.5 dB DRR)
- 2) **Reverberation Variation Group:** Three scenarios were defined with a constant DRR of 6 dB to investigate the independent effect of decay time on perceived quality. For this group, a single female speaker was used, positioned at an azimuth of 60° and an elevation of 90° :
 - Scenario E:** $9.2 \times 13.4 \times 2.2$ m ($T_{60} = 270$ ms)
 - Scenario F:** $16.0 \times 15.0 \times 2.5$ m ($T_{60} = 450$ ms)
 - Scenario G:** $18.0 \times 15.5 \times 3.2$ m ($T_{60} = 750$ ms)

For each scenario, four distinct stimuli were prepared. First, SH-RIRs were generated at three Ambisonics orders: 8th, 3rd, and 1st (FOA). These were convolved with the dry signals to produce the corresponding Ambisonics signals. The fourth stimulus was created by upscaling the FOA signal using the proposed FM framework. Lastly, all signals were rendered binaurally using the least-squares method [54] with

²Audio samples comparing all evaluated methods are available online at <https://amitmls.github.io/GenAU>

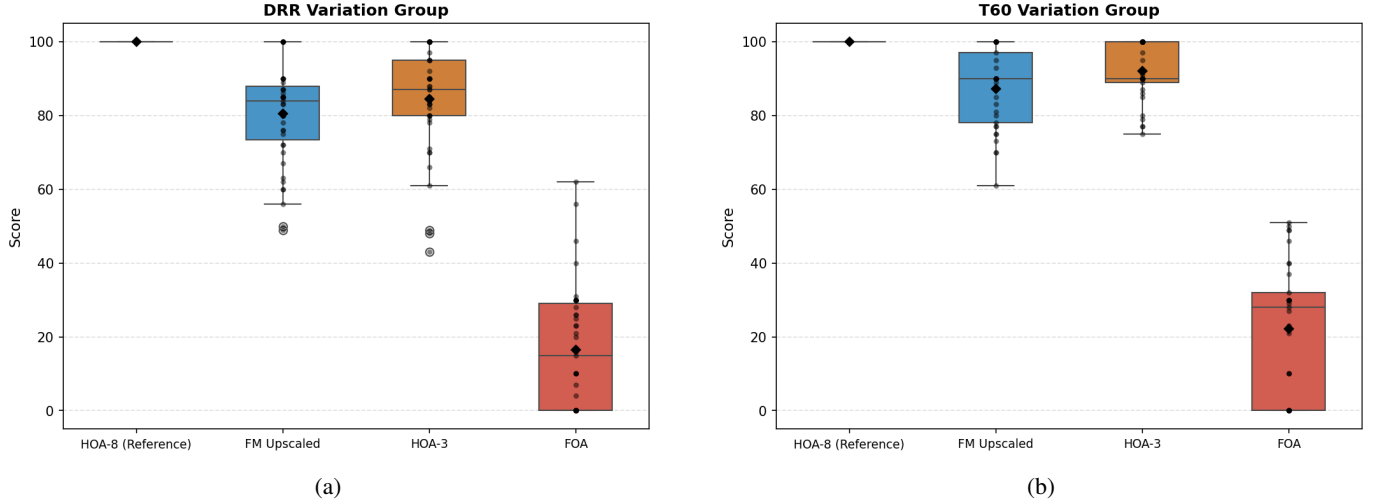


Fig. 6: MUSHRA Results: (a) scores for DRR variation group; and (b) scores for T_{60} variation group

Neumann KU100 HRTFs and normalized to -6 LUFS to ensure perceptual loudness equality across all conditions. To focus the assessment strictly on static spatial accuracy, no head-tracking was employed during the evaluation.

B. Methodology

The subjective evaluation followed a Multiple Stimuli with Hidden Reference and Anchor (MUSHRA) protocol [55]. Participants were tasked with rating the *overall spatial fidelity* of four stimuli compared to a known reference: (i) a hidden 8th-order reference, (ii) a FOA anchor, (iii) an upscaled HOA3 signal using the FM framework, and (iv) the ground-truth HOA3.

The evaluation was conducted by 11 listeners, each of whom participated in a brief training session to become familiar with the range of stimuli and the interface. To mitigate potential order bias, the acoustic scenarios were presented in a randomized sequence.

The participants provided ratings on a continuous scale from 0 to 100, where a score of 100 represents no audible difference from the reference, where lower scores indicate larger differences.

C. Results

The results of both experiments are illustrated in Fig. 6.

1) *DRR Variation Group*: A two-way repeated-measures analysis of variance (ANOVA) was conducted to evaluate the impact of the processing method and DRR levels on subjective quality ratings. The analysis of the main effect of the processing method first revealed that Mauchly's test of sphericity was violated ($\chi^2(5) = 15.00, p = 0.011$); consequently, Greenhouse-Geisser corrections were applied to the degrees of freedom. The results indicate a statistically significant main effect of the method ($F(3, 30) = 149.17, p < .001, \eta_p^2 = 0.94$), suggesting that the processing method significantly influences perceived quality regardless of the acoustic environment.

In contrast, the main effect of the DRR level was not statistically significant ($F(3, 30) = 0.35, p = 0.787, \eta_p^2 = 0.03$), indicating that variations in the DRR within the tested range did not significantly alter the subjective scores. Furthermore, no

significant interaction between the processing method and DRR was observed ($F(9, 72) = 1.86, p = 0.190, \eta_G^2 = 0.06$). This suggests that the performance of the evaluated methods remained robust and consistent across the different DRR scenarios.

Finally, post-hoc pairwise comparisons were performed to evaluate the proposed system against the ground-truth HOA3 across all DRR conditions; no significant differences were found in any of the tested scenarios ($p > 0.05$). Despite the lack of statistical significance, this per-scenario analysis revealed a clear perceptual trend: the gap between the upscaled output and the ground truth narrowed consistently as the DRR increased. At the lowest DRR condition (DRR = -2.5 dB), a mean score difference of 9.4 was observed with a large effect size ($g = 0.82$), favoring the ground truth. As the DRR increased to 0.0 dB and 3.5 dB, the mean score difference reduced to 4.3 ($g = 0.41$) and 1.1 ($g = 0.07$), respectively. Finally, at 9.5 dB, the difference reached -2.09 ($g = -0.11$), indicating that the perceptual gap had effectively closed and slightly reversed in favor of the upscaled output. This convergence in performance across higher DRR scenarios directly correlates with the objective trends observed in Subsection IV-C.

2) *T_{60} Variation Group*: A two-way repeated-measures ANOVA was also conducted to analyze the influence of the processing method and T_{60} levels on subjective quality. Similar to the DRR group, Mauchly's test indicated that the assumption of sphericity was violated for the main effect of the method ($\chi^2(5) = 23.59, p < .001$), requiring Greenhouse-Geisser corrections. The analysis revealed a statistically significant main effect for the processing method ($F(3, 30) = 136.76, p < .001, \eta_p^2 = 0.93$). Conversely, no statistically significant main effect was found for the T_{60} level ($F(2, 20) = 0.66, p = 0.526, \eta_p^2 = 0.06$), and the interaction between method and T_{60} was likewise non-significant ($F(6, 60) = 1.23, p = 0.305, \eta_G^2 = 0.03$).

Post-hoc pairwise comparisons between the proposed system and the HOA3 ground truth further elucidated the relationship between decay time and perceived quality. While the global difference was not statistically significant ($p = 0.298, g = 0.64$), the per-scenario analysis confirmed that no significant differences

were found in any individual scenario ($p > 0.05$). Despite this, a clear perceptual trend emerged: the performance gap widened as the reverberation time increased. Specifically, at the lowest reverberation level ($T_{60} = 0.27$ s), the mean difference score was 1.64 with a small effect size ($g = 0.16$), indicating the upscaled signal was perceptually equal to the ground truth. However, as the T_{60} increased to 0.45 s and 0.75 s, the mean difference score grew to 6.0 ($g = 0.54$) and 6.8 ($g = 0.82$), respectively. This trend suggests that while the proposed method performs excellently in less reverberant conditions, the perceptual difference from the ground truth becomes more pronounced as the effect of reverberation increases (although not statistically significant), a result that aligns with the objective trends observed in Subsection IV-C.

3) *Summary*: While the FM framework outperformed the baseline in the quantitative analysis in Section IV, the listening test results indicate a remaining (although statistically insignificant) perceptual gap between the upscaled output and the ground truth HOA3 in highly reverberant scenarios. Closing this gap remains a primary objective for future work. Nevertheless, we believe these results demonstrate that a generative approach to AU is a highly promising direction, providing a robust theoretical framework for addressing what is inherently a severely underdetermined inverse problem.

VI. LISTENING TEST II - AU METHODS COMPARISON

We conduct an additional listening test to compare different AU methods, spanning model-based, generative, and discriminative data-driven approaches. To this end, Subsections VI-A and VI-B describe the setup and methodology, respectively, while Subsection VI-C presents the results and a discussion of the findings.

A. Setup

We compare single- and multi-speaker settings using the same room as in *Scenario F* ($T_{60} = 450$ ms, DRR = 6dB) from the preceding section. The first scenario evaluated a single speaker positioned at $(\theta, \phi) = (90^\circ, 20^\circ)$, whereas the second evaluated a multi-speaker scene with three concurrent speakers located at $(90^\circ, 30^\circ)$, $(90^\circ, -60^\circ)$, and $(90^\circ, 90^\circ)$, respectively. All signals were generated as described in Subsection V-A.

B. Methodology

We followed the same MUSHRA methodology as in Subsection V-B, with the exception that five stimuli were presented: (i) a hidden HOA3 reference, (ii) a FOA anchor, and 3rd-order HOA signals upscaled from FOA using (iii) the proposed FM framework, (iv) CTN (a deterministic baseline), and (v) PWD CS (a model-based baseline). With the aim of focusing on between-method comparison while keeping the number of stimuli small, the 8th-order Ambisonics reference was omitted. The evaluation was conducted by 11 listeners experienced in spatial audio testing.

C. Results

Results of the method comparison experiment are presented in Fig. 7. A two-way repeated-measures ANOVA was conducted to evaluate the impact of the processing method. Mauchly's test of

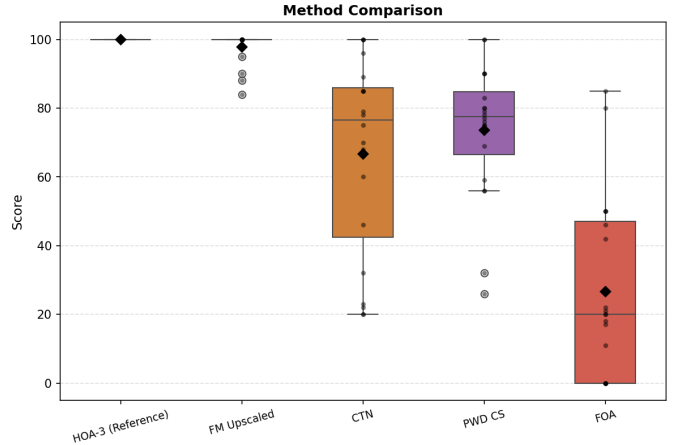


Fig. 7: MUSHRA results for AU method comparison

sphericity was violated ($\chi^2(9) = 37.11, p < .001$); consequently, Greenhouse-Geisser corrections were applied to the degrees of freedom. The results indicate a statistically significant main effect of the method ($F(1.62, 14.56) = 25.02, p < .001, \eta_p^2 = 0.94$), suggesting that the processing method significantly influences perceived quality. As can be seen, CTN scored significantly lower than the proposed method and performed comparably to the model-based baseline, despite outperforming the latter in signal fidelity. Listener feedback indicated that this subjective degradation was driven by intermittent clicking artifacts in the CTN output, which were noticed by a subset of participants. Post-hoc pairwise comparisons with Bonferroni correction between the proposed system and other methods revealed no significant difference compared to ground truth HOA3 ($p = 0.837, g = -0.83$). The comparison to CTN showed a trend toward significance but did not cross the standard threshold ($p = 0.051, g = -1.61$), whereas the difference compared to PWD CS was statistically significant ($p = 0.013, g = -1.73$). These findings demonstrate that the FM method matches the perceptual quality of ground-truth HOA3, while the competing approaches produced noticeable artifacts.

VII. CONCLUSIONS

In this work, we proposed a novel generative framework for AU, formulating the spatial upscaling of reverberant speech as a principled inverse problem. By learning the underlying data manifold, the proposed method provides a robust solution for synthesizing high-order spatial audio across diverse acoustic conditions. Our comprehensive evaluation across varied scenarios demonstrates the clear benefits of this approach for AU. Furthermore, while not perceptually identical to the HOA3 ground truth in all scenarios, the performance gap was not statistically significant.

Our framework requires a larger parameter footprint and higher computational complexity than deterministic baselines, which may limit its immediate real-time utility. However, it successfully demonstrates the strong potential of generative paradigms to capture complex spatial distributions with high fidelity. Accordingly, future work will focus on refining the generative process to preserve nuanced spatial cues, alongside architectural optimization and model reduction to facilitate

real-time deployment. We also intend to explore latent-space formulations to efficiently scale to higher sampling rates.

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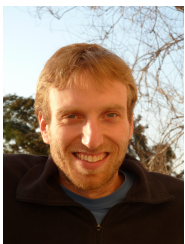
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